Project Diary to be maintained by each student

PROJECT DIARY (STUDENT)

Name: ..... **Enrollment No.:** .....

Project Title: Drivix – Smart Parking & Urban Mobility Platform

Mentor: Prof. Reeti

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
One	10/03/26 12/03/26	Initial resources SPS, IoT concept exploration	Guide suggested focus on real-world parking problem	Prepare detailed problem statement	Signed (Reeti)
Two	15/03/26 18/03/26	Literature survey on smart parking systems	Improve feasibility planning and hardware scope	Continue architecture planning	Signed (Reeti)
Three	20/03/26 22/03/26	Workflow planning completed	Improve model structure & system flowcharts	Study database and API frameworks	Signed (Reeti)
Four	24/03/26 26/03/26	API and database research concluded	Focus on usability & seamless driver experience	Start implementation planning	Signed (Reeti)
Five	28/03/26 31/03/26	Backend & frontend modular planning	Improve module separation (Web vs Mobile app)	Continue API contract planning	Signed (Reeti)
Six	30/03/26 02/04/26	Frontend & backend architectural planning	Focus on scalable backend & concurrent booking locks	Begin implementation preparation	Signed (Reeti)
Seven	07/04/26 09/04/26	QR Navigation workflow defined	Planning appreciated by guide; approved design	Continue implementation planning	Signed (Reeti)
Eight	14/04/26 15/04/26	CORE PHASE Implementation planning finalized; Tech stack modularized	Stack choices approved; Maintain clear separation of concerns	Setup monorepo and initialize backend Express environment	[To Sign]

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PROJECT DIARY (STUDENT) — MONOREPO & BACKEND INFRASTRUCTURE

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
Nine	**21/04/26 22/04/26**	Monorepo structured (`WebApp`, `App`, `backend`); Express server initialized	Good clean structure; Ensure robust database connection security	Design MongoDB Atlas schemas for Users, Hubs, and Slots	[To Sign]
Ten	**28/04/26 29/04/26**	MongoDB Schemas & database connection established	Verified schema design; Focus on secure tokenized authentication	Implement JWT authentication and OTP registration flow	[To Sign]
Eleven	**05/05/26 06/05/26**	JWT authentication & OTP verification endpoints built	Authentication verified; Advised adding rate-limiting on OTP routes	Develop Parking Hub and dynamic Slot inventory management APIs	[To Sign]
Twelve	**12/05/26 13/05/26**	Parking Hub listing & slot status endpoints active	Verified API responses; Ensure multi-floor capacity tracking	Design automated slot allocation logic algorithm	[To Sign]
Thirteen	**19/05/26 20/05/26**	Slot allocation engine (`SlotAllocationService`) formulated	Check for race conditions & double-booking under concurrency	Implement atomic soft-lock concurrency mechanism with timer	[To Sign]
Fourteen	**26/05/26 27/05/26**	5-minute atomic soft-lock engine & auto-release worker running	Effective lock mechanism; Verify background worker under load	Setup Socket.IO server hub for real-time slot occupancy broadcast	[To Sign]
Fifteen	**02/06/26 03/06/26**	Socket.IO real-time hub connected with room subscriptions	Approved real-time architecture; Keep socket payloads minimal	Develop ANPR automated gate entry/exit simulator logic	[To Sign]
Sixteen	**09/06/26 10/06/26**	ANPR Gate Controller (`gateController.js`) simulated with license plates	Gate cycle confirmed; Create visual web simulator dashboard	Build React WebApp ANPR Gate simulator UI dashboard	[To Sign]

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PROJECT DIARY (STUDENT) — MOBILE APP & BOOKING ARCHITECTURE

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
Seventeen	16/06/26 17/06/26	`AnprGateSimulator.jsx` web interface connected to live backend	Demonstration appreciated; Now initiate mobile client setup	Initialize React Native Expo mobile app and theme system	[To Sign]
Eighteen	23/06/26 24/06/26	Mobile repo configured with Expo, navigation stack, & theme tokens	Clean layout foundation; Ensure cohesive colors and spacing	Implement mobile Axios API client with JWT interceptors	[To Sign]
Nineteen	30/06/26 01/07/26	Mobile API client & secure storage token handling completed	Verified state retention; Ready for authentication screens	Create mobile Auth screens (Login, OTP registration)	[To Sign]
Twenty	07/07/26 08/07/26	Mobile authentication & user profile onboarding screens completed	Tested login successfully; Handle keyboard avoidance on mobile	Build Parking Discovery Screen with search and filter tabs	[To Sign]
Twenty One	14/07/26 15/07/26	`ParkingDiscoveryScreen` completed with live availability badges	Search is snappy; Add clear distance and pricing tags	Implement Parking Detail Screen with hub amenities and tariff	[To Sign]
Twenty Two	21/07/26 22/07/26	`ParkingDetailScreen` designed with pricing and facility info	Guide suggested intuitive floor and vehicle type selection	Develop `FloorSelectionScreen` with floor capacity indicators	[To Sign]
Twenty Three	28/07/26 29/07/26	Interactive Floor Selection screen integrated with backend slots	Visual slot layout approved; Prevent selection of locked slots	Implement Booking Duration & timing selection screen	[To Sign]
Twenty Four	04/08/26 05/08/26	`DurationSelectionScreen` with hourly/daily pricing implemented	Tariff logic verified; Provide clear booking summary before checkout	Create Booking Review screen and booking context state	[To Sign]

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PROJECT DIARY (STUDENT) — NAVIGATION, MAP UX & MOBILITY SERVICES

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
Twenty Five	**11/08/26 12/08/26**	`BookingReviewScreen` & active reservation state management completed	Advised generating unique digital QR pass upon confirmation	Implement Active Parking session screen with live countdown timer	[To Sign]
Twenty Six	**18/08/26 19/08/26**	`ActiveParkingScreen` with session QR pass & timer built	QR entry pass verified; Plan user vehicle vault and history	Implement Booking History screen and vehicle profile management	[To Sign]
Twenty Seven	**25/08/26 26/08/26**	Profile Screen, document vault, and booking history screens done	User profile complete; Proceed to interactive map-based discovery	Build interactive Map Modal with hub markers and geo-coordinates	[To Sign]
Twenty Eight	**01/09/26 02/09/26**	`NetworkMapModal` developed with map pin filtering	Bottom sheet obstructs map; Suggested compact floating card	Replace bulky map overlay with centered floating popover card	[To Sign]
Twenty Nine	**08/09/26 09/09/26**	Compact floating popover card integrated on Web and Mobile map	Map visibility and responsiveness greatly improved; Add live routing	Implement direct live GPS turn-by-turn navigation (Uber/Rapido style)	[To Sign]
Thirty	**15/09/26 16/09/26**	`launchLiveGpsNavigation` integrated with native device maps	Routing tested smoothly; Expand platform to driver care services	Build Car Wash & Detailing discovery screen and backend routes	[To Sign]
Thirty One	**22/09/26 23/09/26**	Car wash center schema, seed utilities, & discovery filter added	Advised replacing generic dummy data with real NCR facility records	Populate real NCR detailing centers via OpenStreetMap script	[To Sign]
Thirty Two	**29/09/26 30/09/26**	Real OpenStreetMap car spa dataset integrated into discovery screen	Real-world data appreciated; Streamline mobile home screen	Streamline Mobile HomeScreen into focused driver dashboard	[To Sign]

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PROJECT DIARY (STUDENT) — POLISH, CLEANUP & FINAL REVIEW

Week	Date of Visit	Status of Task assigned during last visit	Remarks	Task to be carried out till next visit	Signature of Guide
Thirty Three	06/10/26 07/10/26	HomeScreen decluttered; Clean 6-icon grid, vehicle status & active session	Clean UI praised; Eliminate synthetic mock data across all modules	Remove all mock/fallback data and enforce clean empty states	[To Sign]
Thirty Four	13/10/26 (Current Visit)	CURRENT REVIEW Completely purged synthetic mock data; Verified live endpoints and graceful empty states	Commended high engineering standards and production readiness; Prepare testing report	Conduct end-to-end integration testing and document test cases	[To Sign]

KEY MILESTONE SUMMARY (FOR VIVA & MENTOR VERIFICATION):

1. Core Infrastructure & Backend:

- Monorepo architecture (`WebApp`, `App`, `backend`).
- MongoDB Atlas cluster with JWT/OTP authentication.
- Real-time multi-client communication using Socket.IO.

2. Smart Allocation & Concurrency:

- Multi-criteria automated slot allocation engine.
- 5-minute atomic soft-lock hold timer preventing double booking.
- ANPR barrier gate simulator with automated optical recognition.

3. Mobile Client (React Native & Expo):

- End-to-end booking flow: Discovery → Detail → Floor → Duration → Confirmation.
- Active parking pass with animated countdown and digital QR gate pass.
- Direct live GPS turn-by-turn routing via Google Maps / Apple Maps.

4. Value-Added Mobility Ecosystem:

- Car Wash & Detailing center discovery with OpenStreetMap NCR data.
- Production polish: zero synthetic mock fallbacks with clean empty states.

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